Copper Demand Forecasting and Supply Risk Analysis in China

SAUDI ARAMCO & Tsinghua University

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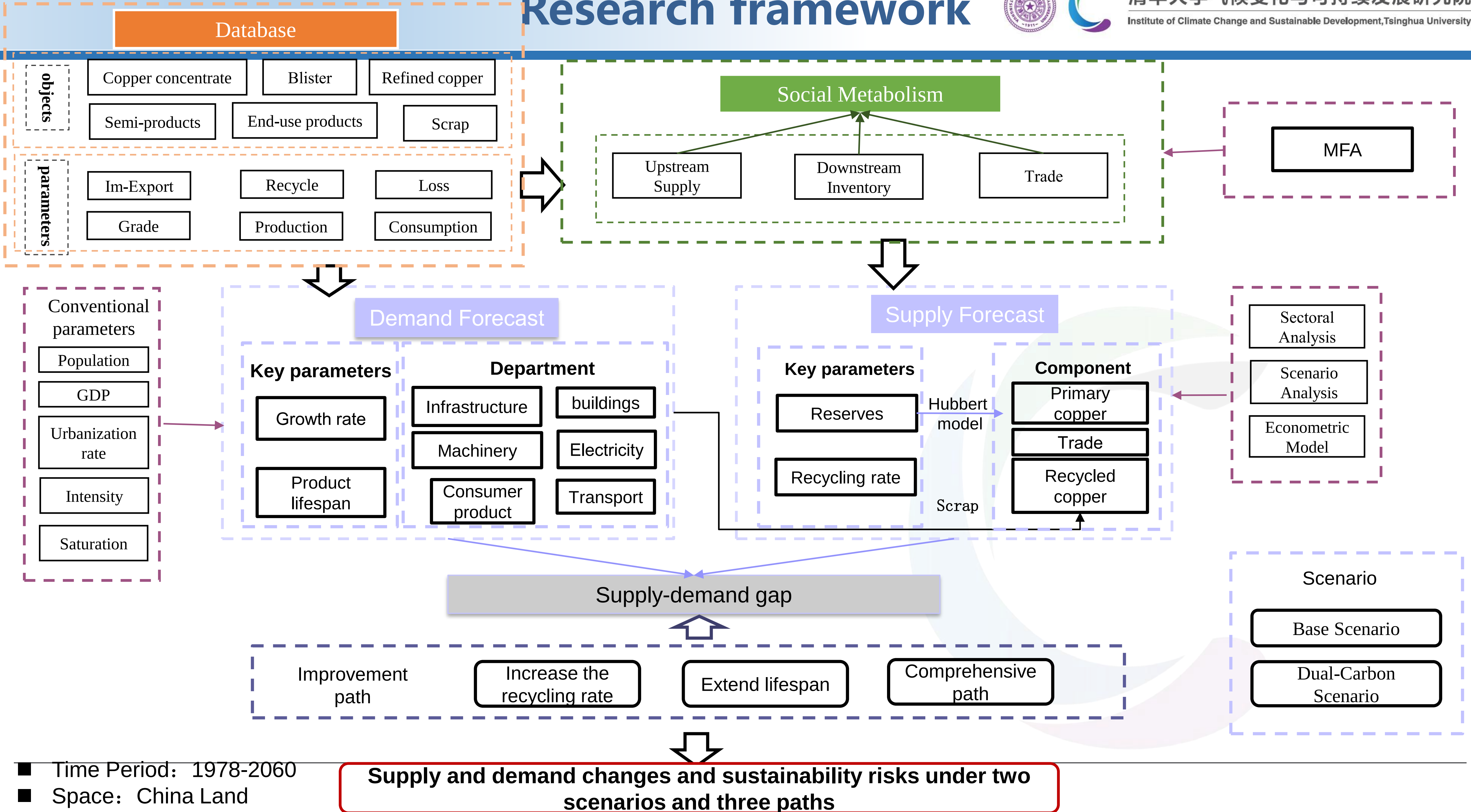
Project Timeline

	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar
Research Proposal & preparations								
Data requirement & database investigation								
Material flow analysis & In-use stocks								
Demand forecasting model & results								
Relationship of supply and demand & Supply risk								

Research framework



- Time Period：1978-2060
- Space：China Land

Demand Forecasting Framework

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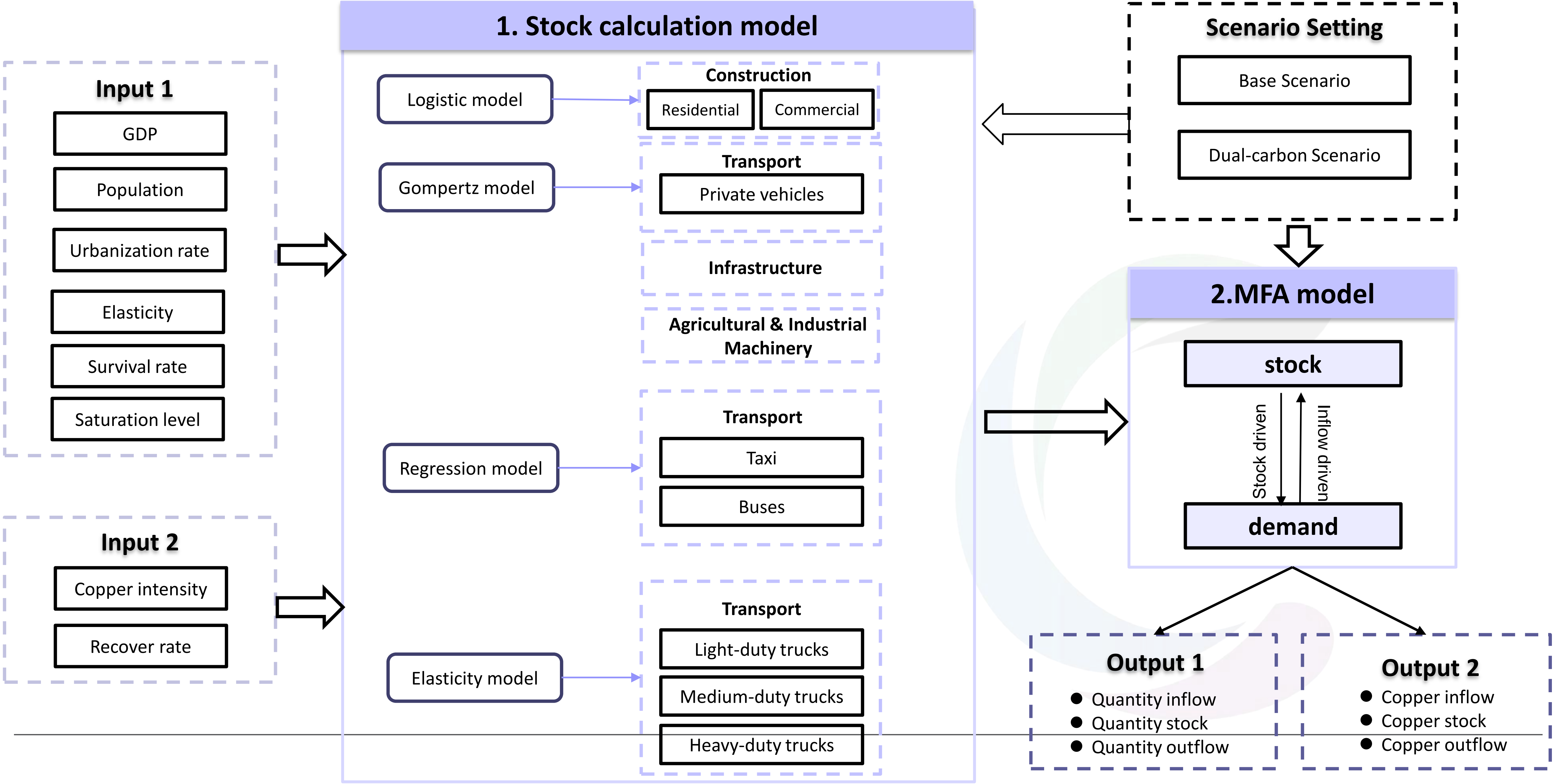
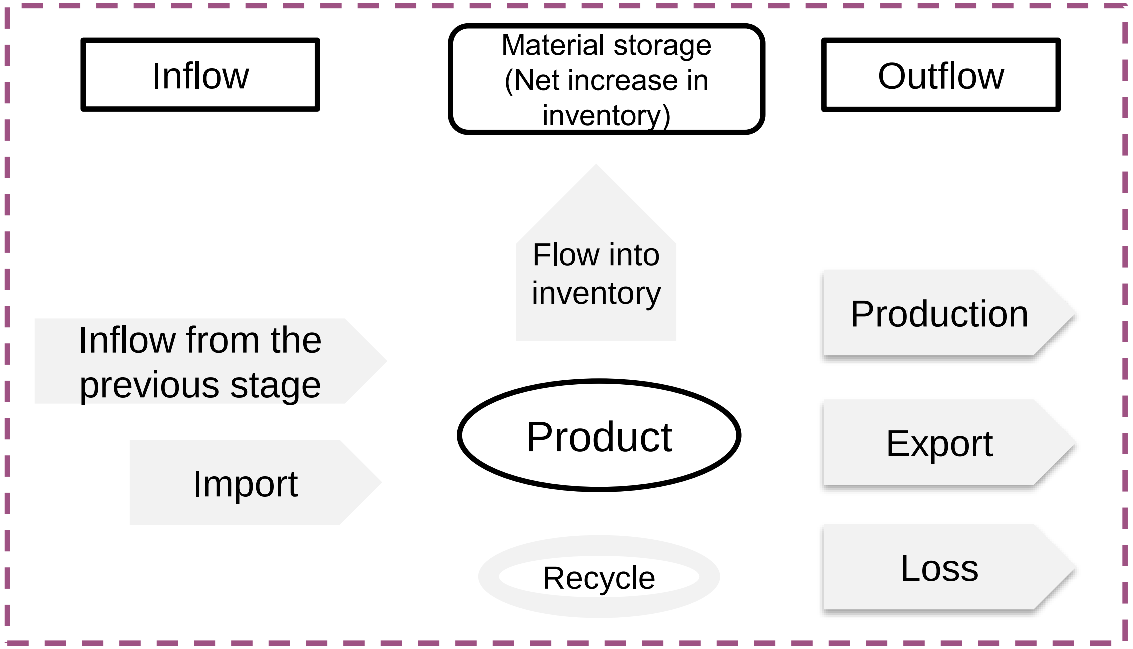


Material Flow Analysis

Material flow analysis is an analytical method that quantitatively tracks the flow characteristics of materials throughout the entire process from extraction, processing, usage to disposal. By establishing the system boundaries for material input, transformation, and output, it calculates the material flux balance relationships within a specific region or industrial system.



$$\Delta \mathrm{Stocks} = \sum \mathrm{F}_{\mathrm{inflow}} - \sum \mathrm{F}_{\mathrm{outflow}}$$

Stock

$$Stocks(t) = \sum_i \underbrace{N_i(t)}_{\text{Year t, Number of product i}} \times \underbrace{m_i(t)}_{\text{Copper intensity of product i}}$$

Outflow（Scrap）

$$\mathrm{Outflow}(t) = \sum_{t'=t0}^{t'=t-1} Inflow_{i,t} \times \left(\frac{1}{\sigma_i \sqrt{2\pi}} e^{-\left(\frac{1}{2}\right) \times \left(\frac{t-t'-\tau_i}{\sigma_i}\right)^2} \right)$$

Inflow（Demand）

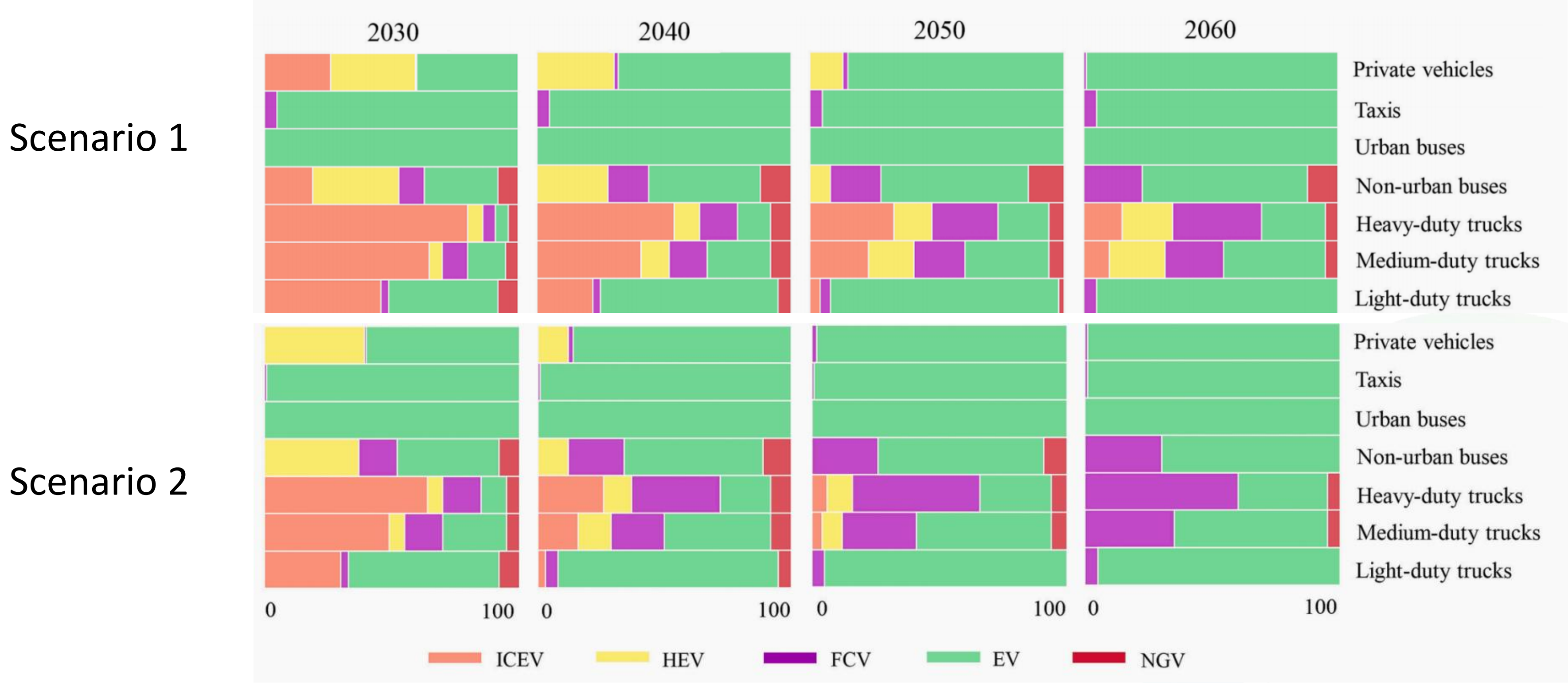
$$Inflow_{i,t} = Stock_{i,t} - Stock_{i,t-1} + Outflow_{i,t}$$

Scenario Setting

Scenario Name	Scenario Introduction
Baseline scenario	Based on the current development situation, projections are made. ■ In the electricity industry, before 2060, thermal power generation and traditional internal combustion engine power generation methods will not be completely phased out; ■ In the road transportation industry, the annual growth rate of new energy passenger cars, new energy buses, and new energy mini-light trucks is determined based on the average growth rate of the past few years. ■ In the construction industry， the copper strength will increase according to the current trend.
Dual-carbon scenario	Based on the newly released new energy plans that are in line with the dual-carbon goals, and considering the rapid development of clean energy technologies. ■ In the electricity industry, by 2060, the installed capacity of clean energy facilities will increase successively in the order of solar energy, wind energy, hydropower and nuclear energy, while thermal power will significantly decrease. ■ In the road transportation industry, the accelerated popularization of new energy vehicles will prompt the replacement of traditional vehicles to be completed in advance. ■ In the construction industry, due to the accelerated electrification process, the growth rate of copper strength will be extremely fast, higher than the previous two scenarios.

Scenario Setting – Road Transport Industry

The proportion of vehicle models in different scenarios



- From 583,200 units in 2015 to 31.4 million units in 2024, a growth of over 50 times.
- Baseline scenario: Assuming that from 2030 onwards, all new sales of light-duty passenger cars and urban buses will be electric or fuel cell vehicles. However, the decarbonization of heavy and medium-sized trucks is more difficult compared to small cars. It is set that by 2060, hybrid vehicles will still account for 15% of the sales of heavy and medium-sized trucks.
- Dual carbon scenario: Refer to the 2050 global net-zero emissions roadmap of the IEA. By 2030, internal combustion engine vehicles will be phased out from the Chinese passenger vehicle sales market, and new sales of light vehicles and urban buses will be fully electrified; In 2060, internal combustion engine vehicles and hybrid vehicles will respectively exit the medium-sized and heavy-duty truck sales markets.

Scenario Setting – Electricity Generation Industry

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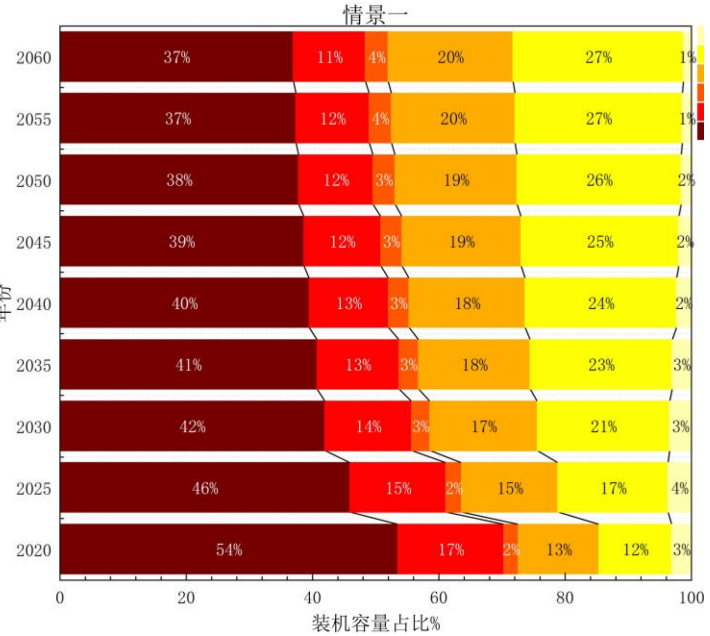
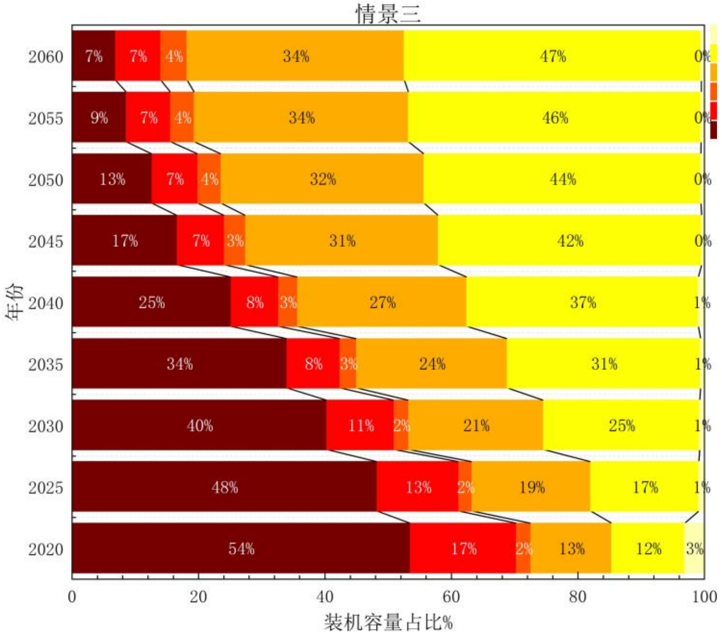
Installation proportion in different scenarios



Scenario 1



Scenario 2

其他
太阳能
风能
核能
水能&水力发电蓄能
化石燃料

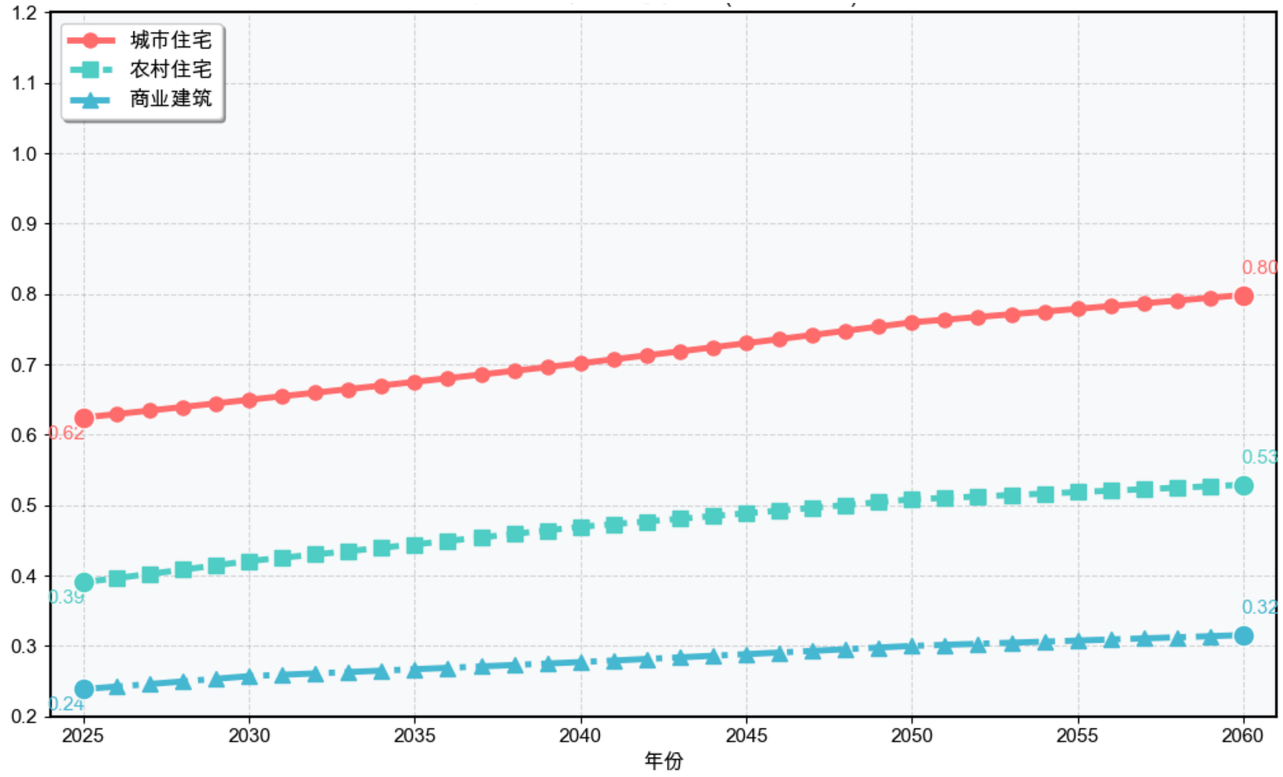
Others
Solar
Wind
Nuclear
Hydro & Hydroelectric Pumped Storage
Fossil Fuels (coal oil gas)

Total installed capacity in different scenarios

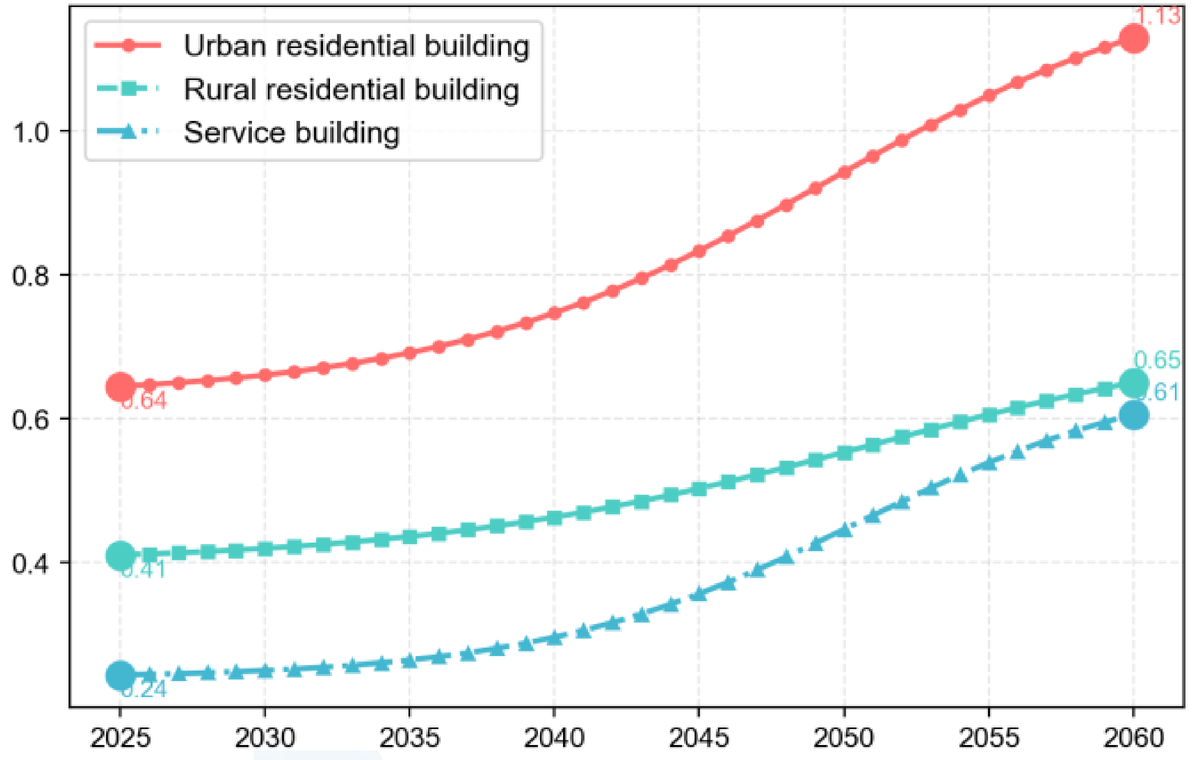
Period	Base Scenario	Dual Carbon Scenario
2025–2030	3.5%	5.9%
2030–2035	2.5%	5.9%
2035–2040	2.5%	3.2%
2040–2045	1.5%	2.6%
2045–2050	1.5%	1.8%
2050–2055	0.8%	1.3%
2055–2060	0.8%	0.7%

Scenario Setting – Construction Industry

Copper intensity in Scenario 1（kg/m^ 2）



Copper intensity in Scenario 2（kg/m^ 2）



- In this study, the total building area of the entire society and its composition (such as the proportions of residential buildings, public buildings, and commercial buildings) remained unchanged under different scenarios. The core parameter that changed under different scenarios was only the copper usage intensity per unit area of buildings.
- In Base scenario, the increase in copper strength per unit area is relatively moderate.
- Under the dual-carbon scenario, the growth follows an S-shaped curve. the copper strength per unit area and its growth rate will be significantly higher than in the baseline scenario.

Results – Road Transport Industry

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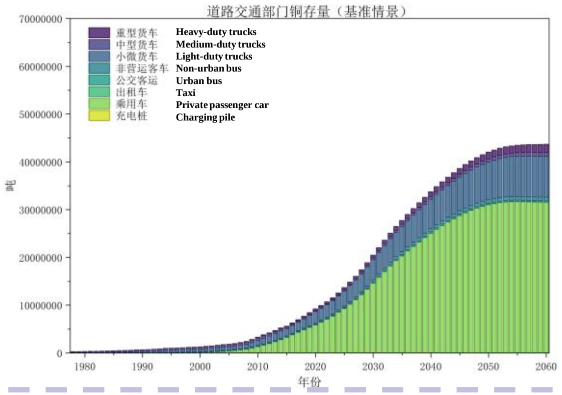
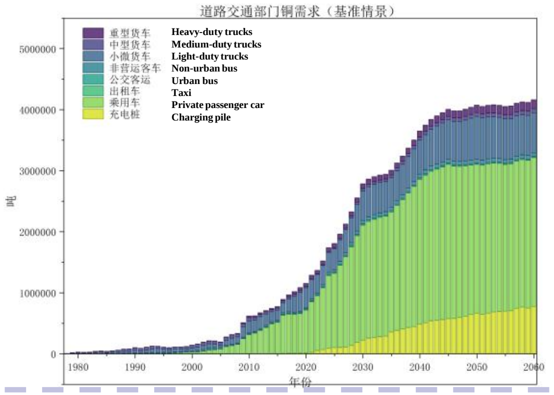
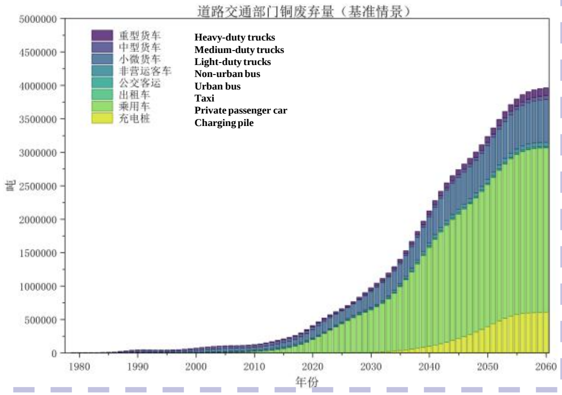
Stock | Demand | Outflow

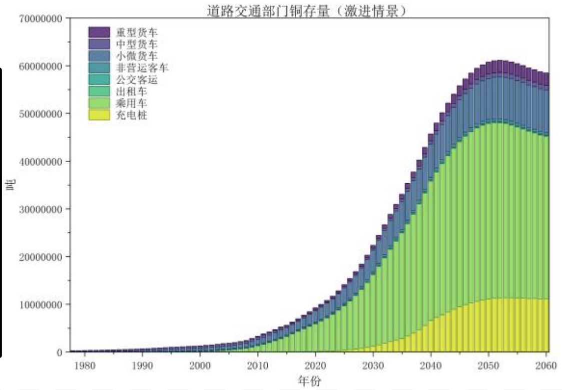
Scenario 1

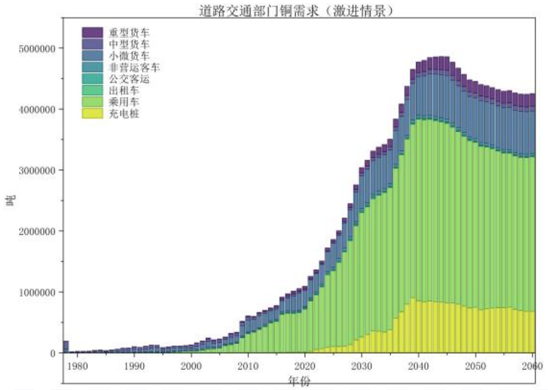


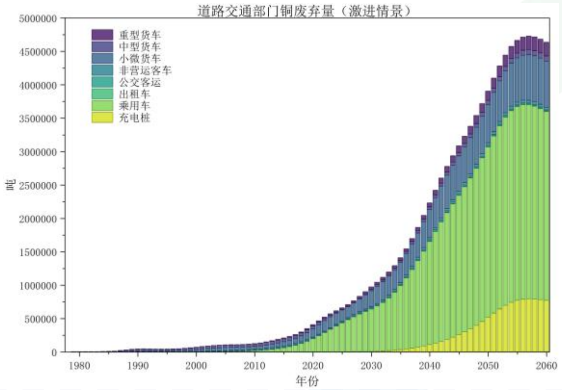




Scenario 2







Copper Intensity（kg/unit）

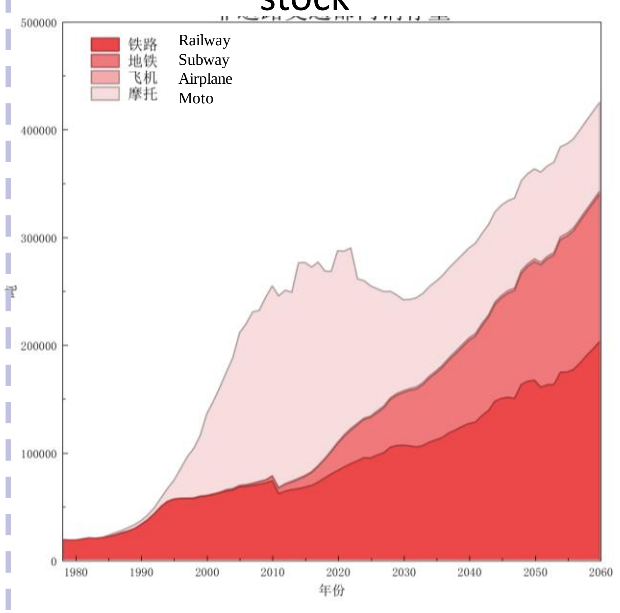
Type	Conventional Vehicles	NEV
Private Passenger Car	25	85
Taxi	25	85
Urban bus	110	300
Non urban bus	110	300
Heavy duty trucks	220	900
Medium duty trucks	200	720
Light duty trucks	40	250

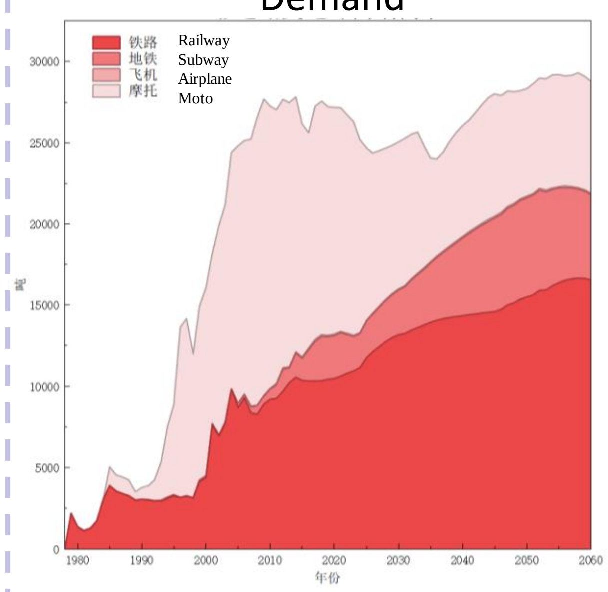
- Baseline scenario: The growth rate of copper demand is relatively slow. The peak of copper demand is reached at 41.6 million tons in 2060.
- Dual Carbon scenario: There is an initial rapid growth followed by a decline in demand. The total demand reaches a peak of 48.6 million tons in 2045. There is a slight decline from 2045 to 2060, and finally it drops to 42.5 million tons in 2060.
- The structural differences in vehicle models are the main cause of the differences in the two scenarios. The gap between two scenario continued to widen and reached its maximum of 30.7% in 2040. The peak in the dual-carbon scenario was not only nearly 800,000 tons higher than that in Scenario One, increasing by approximately 19.6%, but also reached earlier by 15 years.

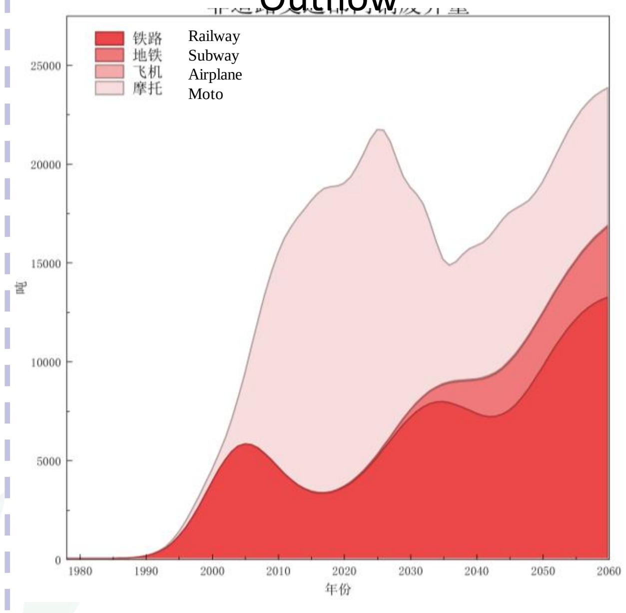
Results — General Department

	stock	Demand	Outflow

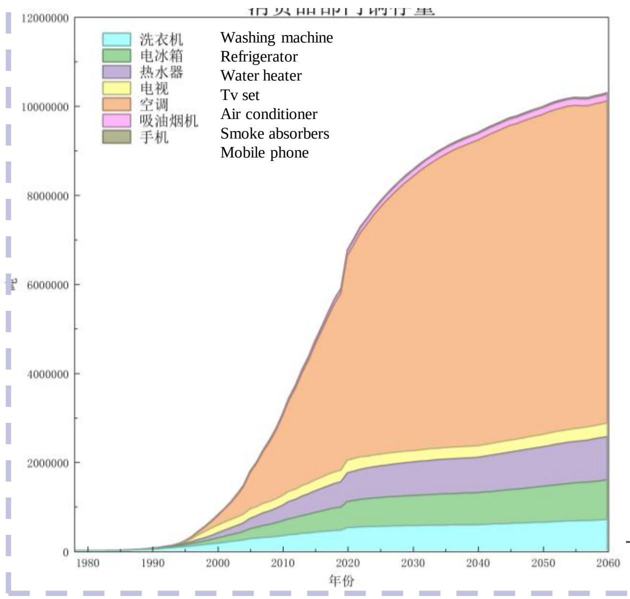
Non road transport

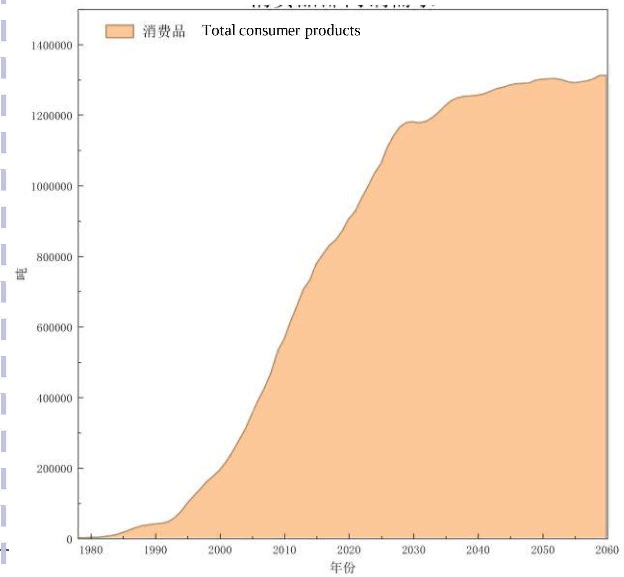


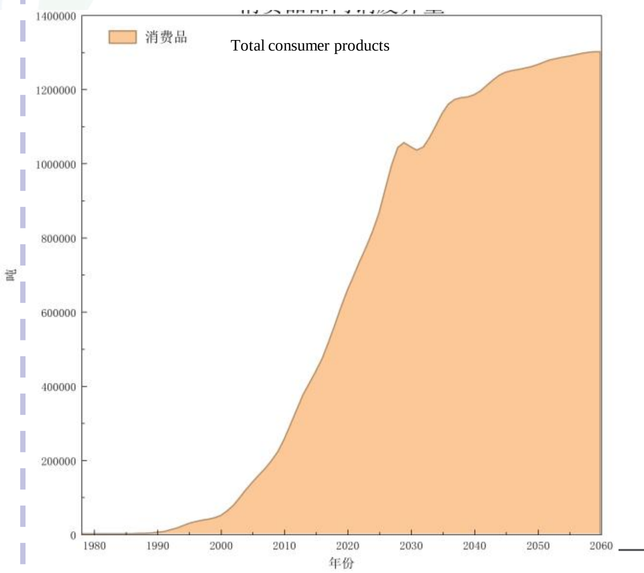




Consumer Products







Results — General Department

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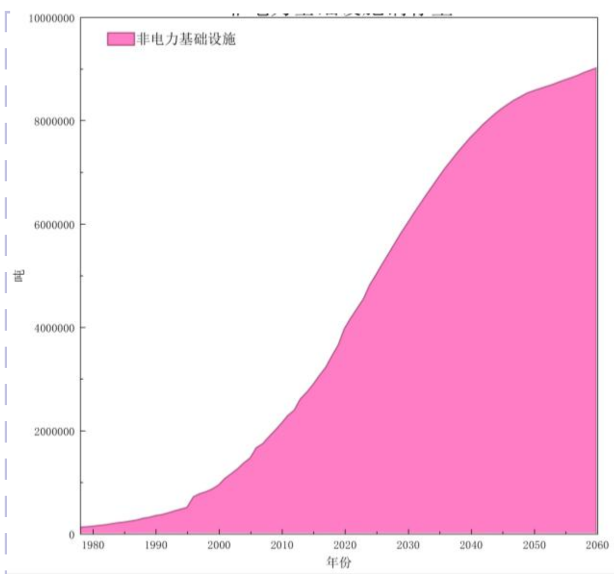
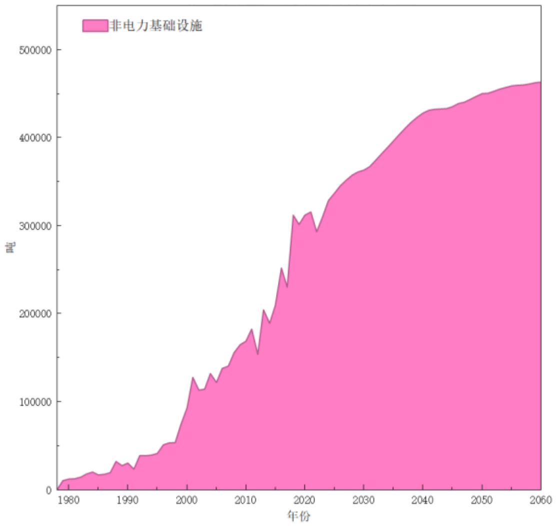
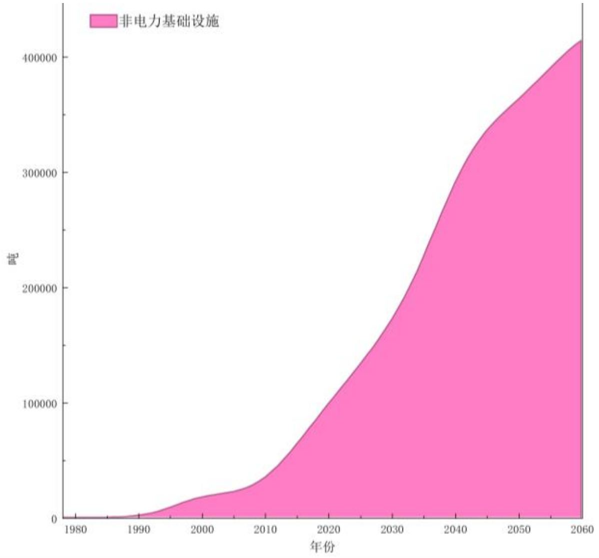
stock | Demand | Outflow

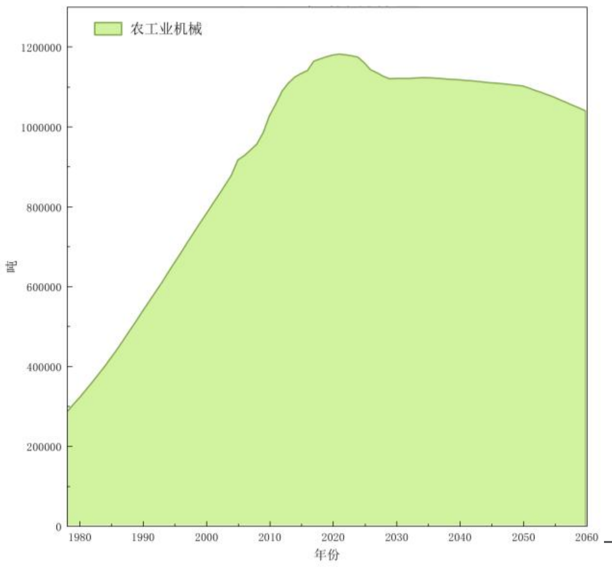
Non - electricity Infrastructure

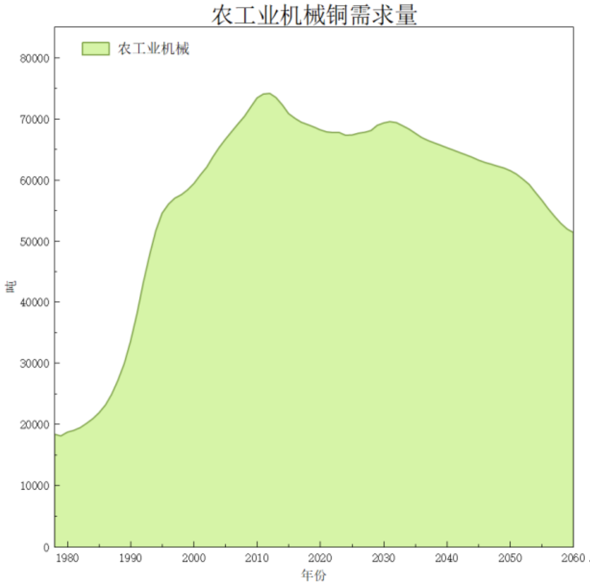


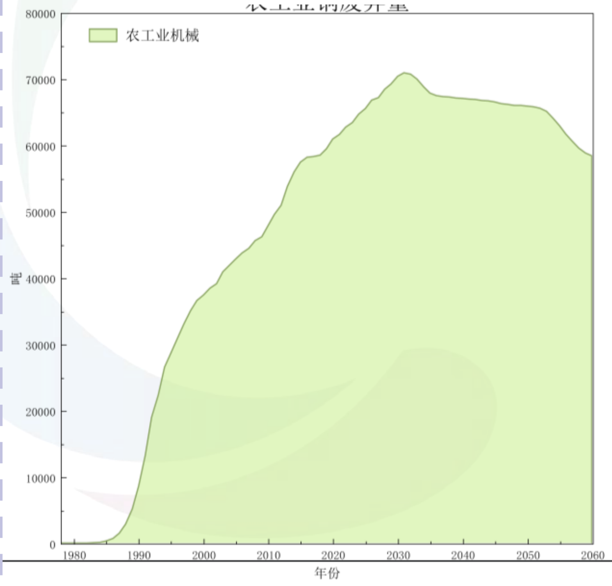




Agriculture & Industrial Products







Results – Electricity Industry

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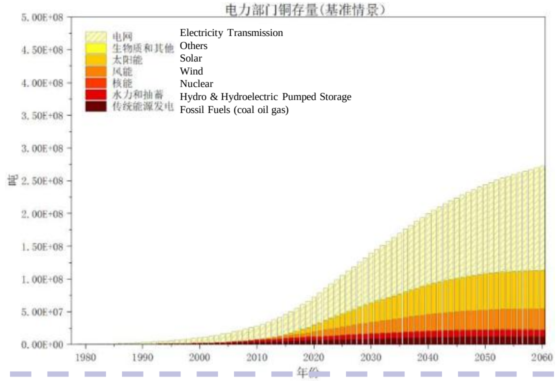
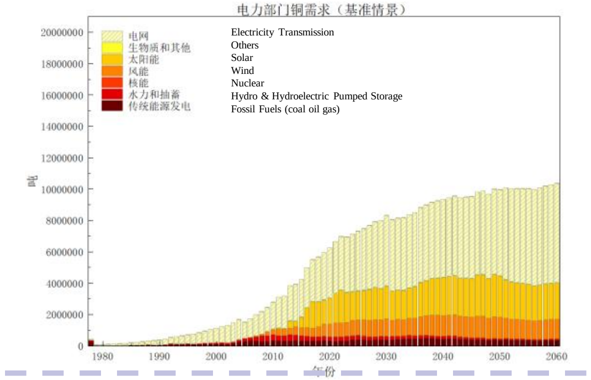
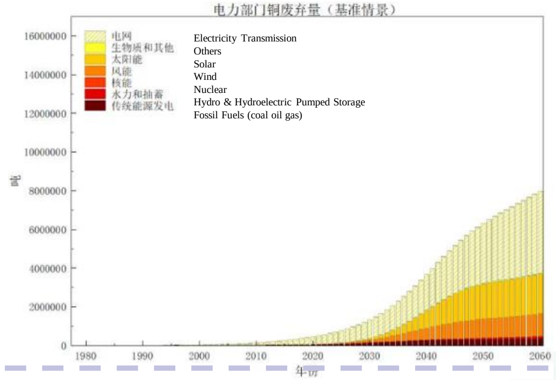
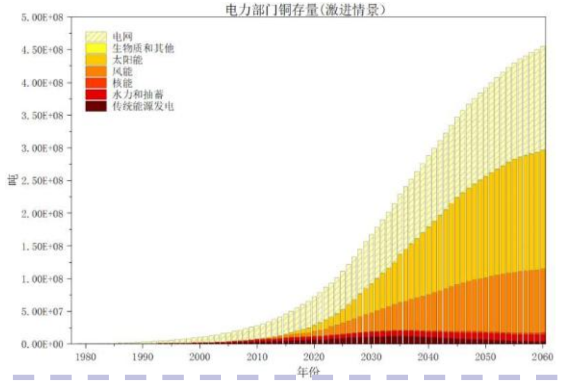
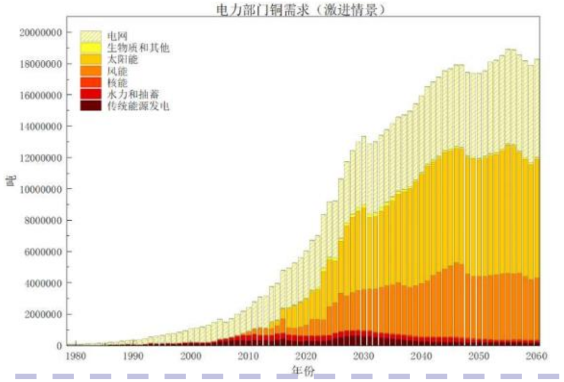
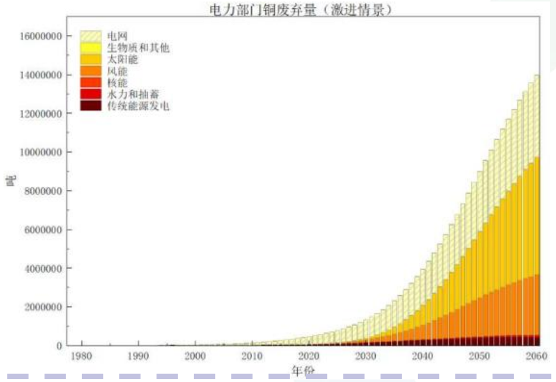
	Stock	Demand	Outflow
Scenario 1			
Scenario 2			

Copper intensity：kg/kw	
Fossil Fuels (coal oil gas)	0.675
Biofuels and waste	1.5
Nuclear	0.75
Hydro	1.75
Geothermal	1.5
Solar	4.5
Wind	3.3
Hydroelectric Pumped Storage	1.75

- Baseline scenario: The copper demand for power generation facilities tends to stabilize as the growth rate of installed capacity slows down. The total copper demand of the power sector increased from 88,000 tons in 1980 to 6.24 million tons in 2020. After that, the growth rate became relatively stable, with an average annual growth rate of only 1.4% from 2020 to 2060, and reached 10.3 million tons in 2060.
- Dual-carbon scenario: There will be rapid growth in the early stage, followed by a slower growth rate and a stable state. It is expected to reach a historical peak of approximately 18.9 million tons around 2055, which is 83% higher than the level in 2060 under the baseline scenario. The average annual growth rate during the peak period from 2020 to 2055 is as high as 4.2%, which is three times that of the baseline scenario during the same period.
- The peak total demand for copper under the dual-carbon scenario will be 189 million tons in 2055; under the baseline scenario, it will peak at 103 million tons in 2060. During the period from 2040 to 2060, the average annual demand level under the dual-carbon scenario is 60% - 85% higher than that under the baseline scenario.

Results – Construction Industry

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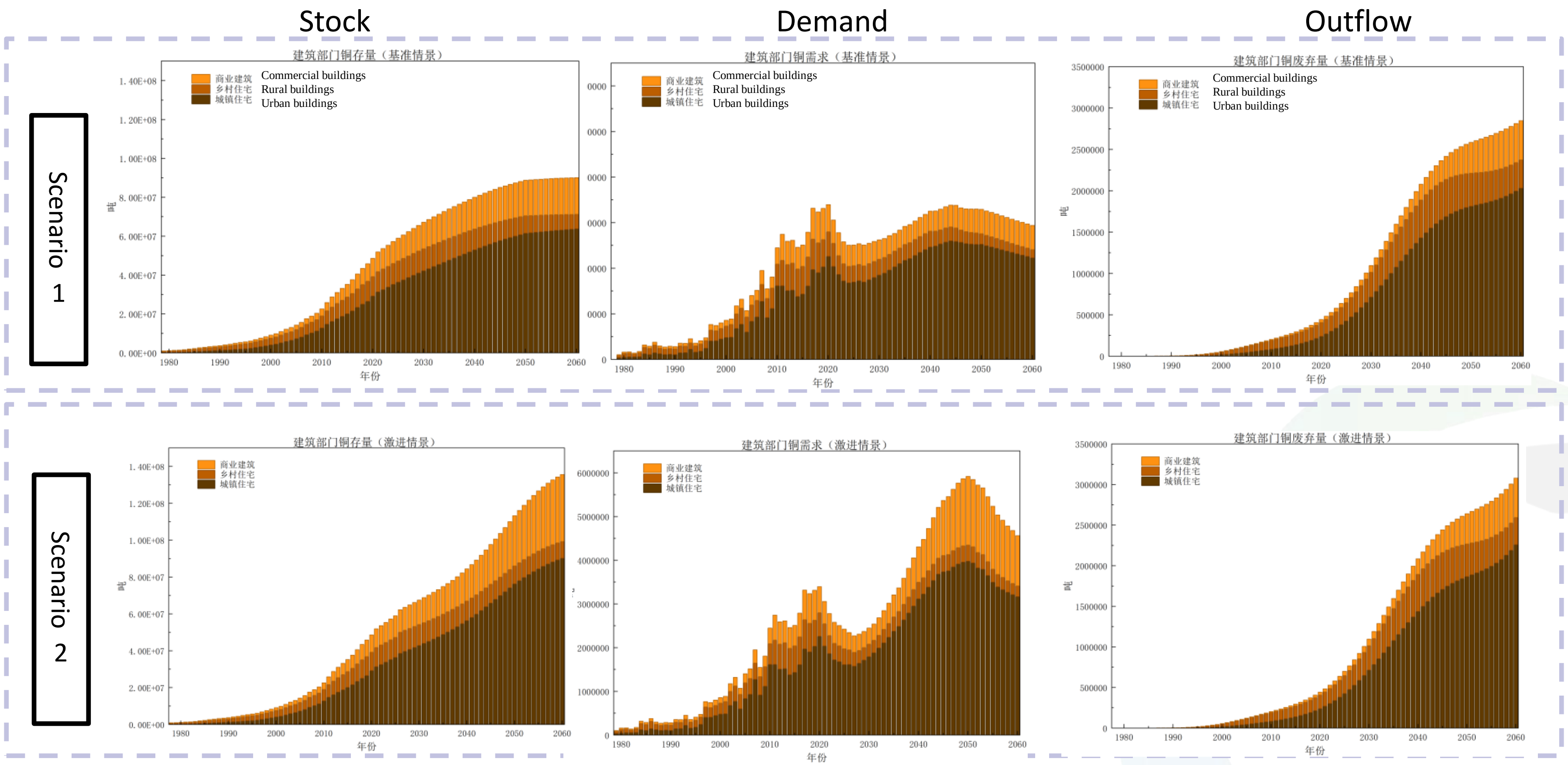
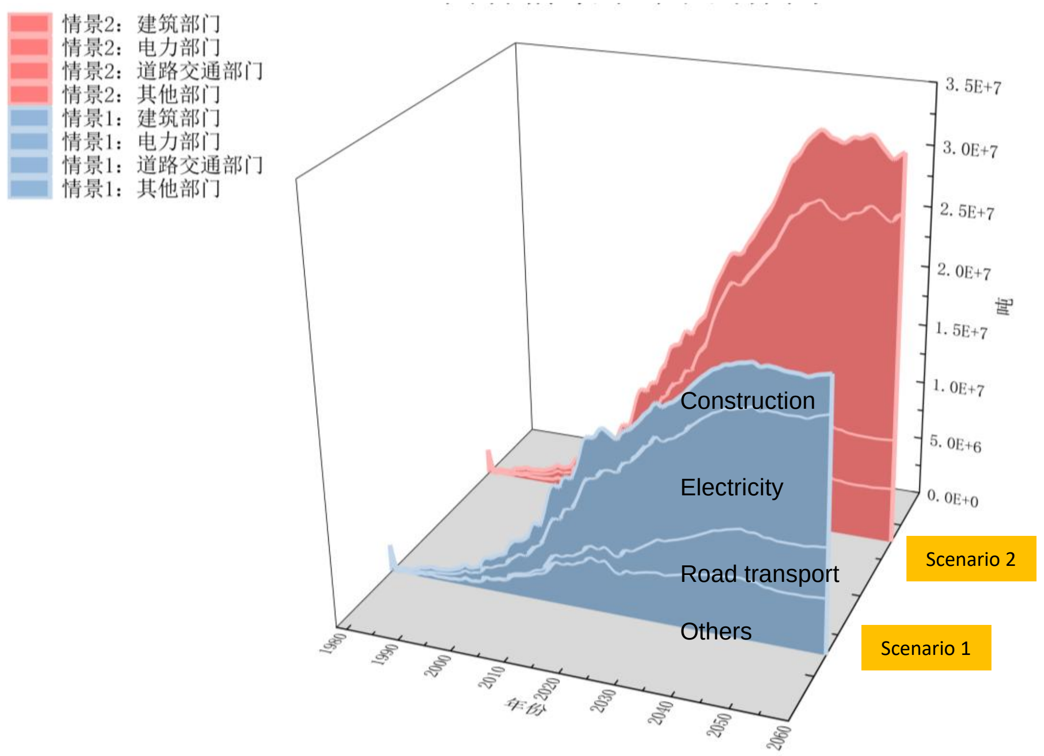
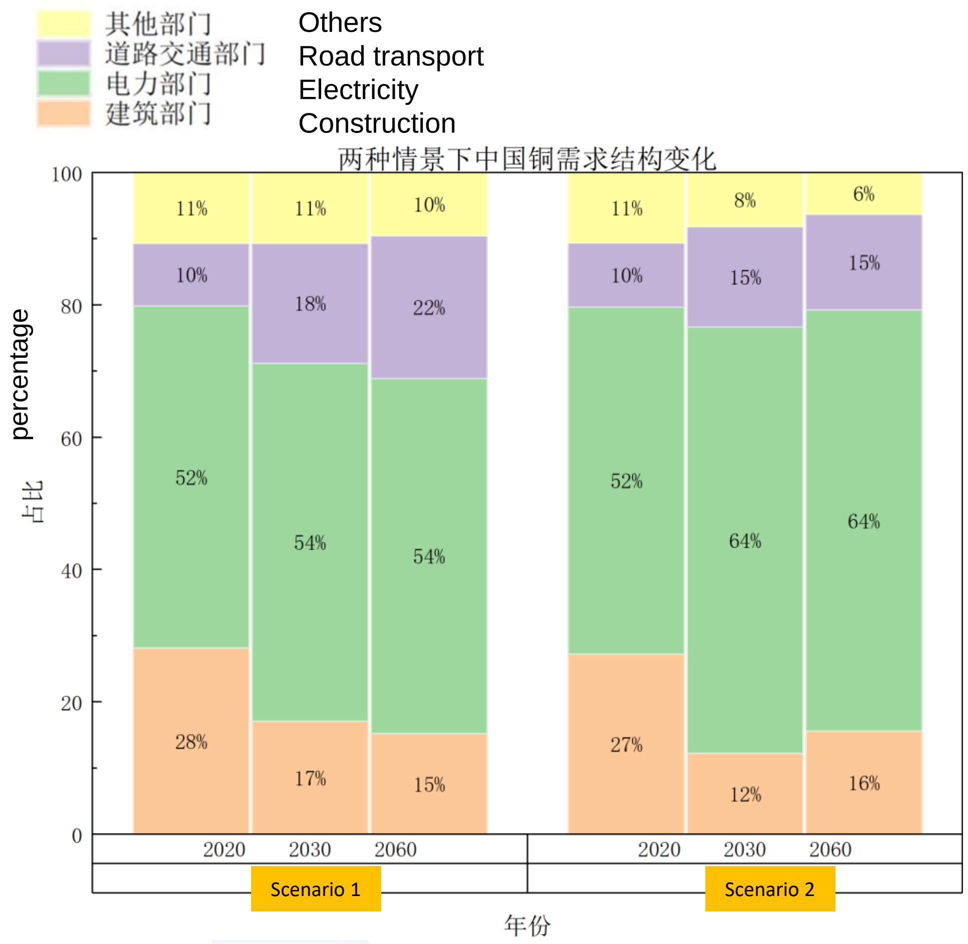


- Under the baseline scenario, the total copper demand of the construction sector follows a curve of peak - decline - re-growth - re-decline, with a relatively smooth transition. The total demand get a peak of 3.4 million tons around 2020. It then showed a trend of rising first and then declining, reaching a peak of 3.99 million tons in 2044 and dropping to 2.94 million tons in 2060.
- Under the dual-carbon scenario, after reaching a peak in 2020, the demand declined briefly and then continued to rise, forming a second peak in 2050, reaching a peak of 5.92 million tons, and falling to 4.57 million tons in 2060.
- Although the construction sector is becoming saturated, due to the increase in copper strength per unit area, the peak demand for copper in the dual-carbon scenario occurs later than in the baseline scenario. By 2035, this gap expands to approximately 10%, and by 2050, it expands to approximately 79.8%.

Results — Comprehensive comparison of scenarios

Copper demands of two scenario

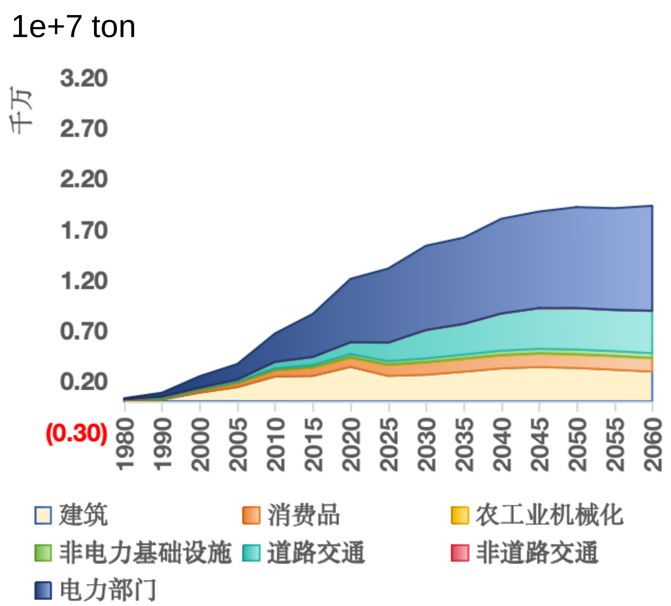




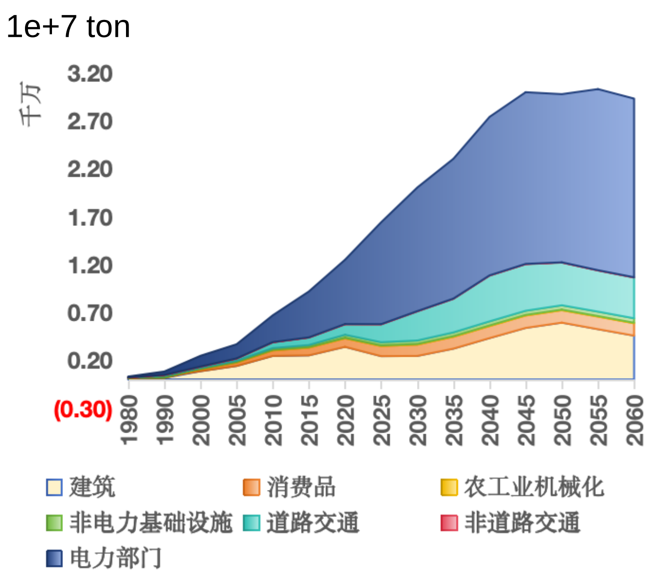
- In the two scenarios, the total demand for the end-use sector in China in 2060 will be 19.3 million tons and 29.3 million tons respectively (↑ 51.8%).
- The proportion of the electricity sector will increase from 52% in 2020 to 54% and 64% in 2060, becoming the main driver of demand growth.
- The proportion of the vehicle transportation sector will increase from 10% in 2020 to 22% and 15%. Due to the squeeze from the power sector, the growth in proportion is limited, but it remains an important driver of copper demand.
- The proportion of the construction sector will decrease from 28% to 15% and 16%. Due to the saturation of construction and the squeeze from the power and transportation industries, its proportion has decreased significantly.

Results – Comprehensive comparison of scenarios

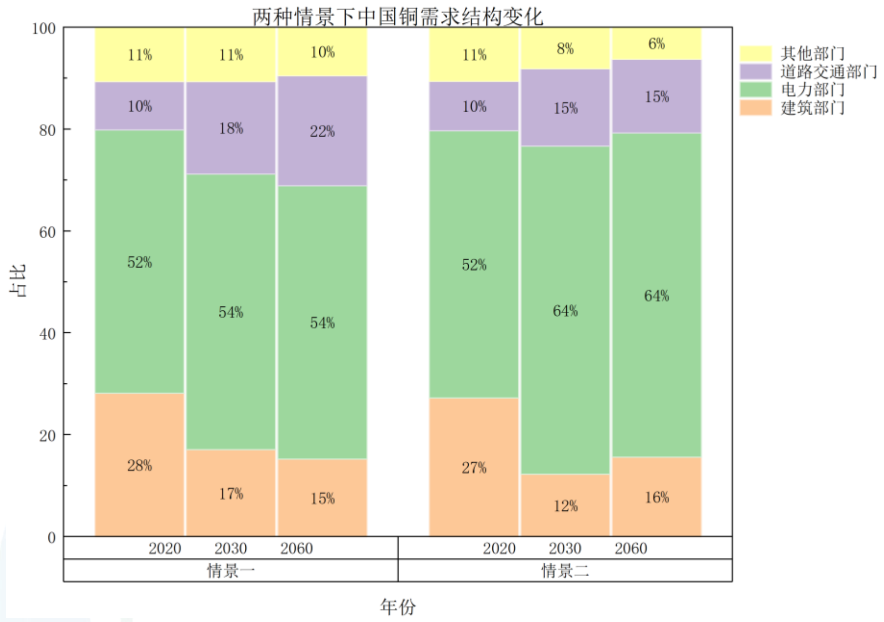
Total Demand of Scenario 1



Total Demand of Scenario 2



Demand Structure of Different Scenario



- Under the base scenario, China's total copper demand will experience a process of steady growth and then gradually reach its peak. The will peak at 192 million tons in 2060. The average annual growth rate from 2020 to 2050 will be only 1.5%. This reflects the general rule that with the economic maturity and the slowdown of the urbanization process, the demand for bulk commodities tends to reach saturation under the traditional development model.
- Under the dual-carbon scenario, the total demand follows a curve that initially surges rapidly and then fluctuates at a high level. In this scenario, the demand reaches a historical peak of 30.53 million tons in 2054, which is approximately 59% higher than the peak level in the baseline scenario. After 2045, the demand shows slight fluctuations, but remains at 29.3 million tons until 2060.
- With the rapid increase in the penetration rates of renewable energy installations and electric vehicles, the total demand gap between the two scenarios gradually widened. It expanded from 25.1% in 2025 to 60.0% in 2045. From 2050 to 2060, the copper gap slightly narrowed after reaching its peak.

Project Timeline and expected output

	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar
Research Proposal & preparations								
Data requirement & database investigation								
Material flow analysis & In-use stocks								
Demand forecasting model & results								
Relationship of supply and demand & Supply risk								

- **Next steps**
  - **Future Supply**
  - **Demand in datacenter**
  - **Deliverables： Model, Report, Database**

Thank You For Your Attention!

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